TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Flash LIDAR for lunar exploration, rover, spacecraft navigation and in-orbit servicing**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Satellite / Radio Wave	Satellite Navigation	Aerospace Systems	Other Aerospace System Technologies
Concept of Proposed Technology	The objective is to demonstrate that a space-qualified flash LiDAR system can deliver robust, full-field 3D perception to support several space missions: 1) lunar descent and landing phases as well as surface exploration and operational activities, 2) spacecraft relative navigation for in-orbit servicing and other applications.			
Rationale for Joint R&D Project	While CSEM has developed a compact flash LiDAR sensor suitable for space applications, real-time 3D reconstruction and perception remain computationally intensive. Addressing these challenges requires the development of advanced processing architectures and optimized algorithms, making a joint R&D effort with specialized Korean SMEs highly relevant.			
Key Words	1. Precision altimetry & topographic mapping – Enables high-resolution 3D surface reconstruction 2. Autonomous navigation / hazard avoidance – Supports GN&C (Guidance, Navigation, and Control) by providing real-time 3D terrain data 3. Real-time 3D sensing – Enables rapid depth perception for applications like autonomous vehicles, robotics, and industrial inspection. 4. Data fusion – data processing optimization by combining RGB or IR camera images with lidar measurements. 5. Radiation-hardened system – supports harsh environmental constraints of space or lunar applications.			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 5	End of the Project	Level 6
R&D Objectives	Enable real-time terrain perception to support Guidance, Navigation, and Control (GN&C) functions during lunar descent and landing, including Terrain Relative Navigation (TRN) and Hazard Detection and Avoidance (HDA). Provide accurate and reliable 3D surface characterization to support In-Situ Resource Utilization (ISRU) and surface operations.			
R&D Target Specifications	Real-time data output suitable for closed-loop GN&C and autonomous rover navigation. Dust-tolerant perception capable of operating during plume-induced dust disturbances in terminal descent. Operational range: from tens of meters (terminal descent) to several tens of meters (rover navigation and surface mapping). Frame rate / acquisition rate: compatible with descent dynamics and rover motion to support real-time decision-making.			
R&D Plan	Phase 1 – Requirements Definition: Consolidation of mission-level requirements for lunar landing or rover operations; and definition of interfaces with GN&C systems. Phase 2 – Technology Development and Prototyping: Implementation and optimization of Flash LiDAR signal processing algorithms for terrain reconstruction, hazard detection, and navigation support, followed by functional and performance testing. Phase 3 – Validation: Closed-loop demonstrations with simulated GN&C and performance assessment.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Overall system architecture definition based on mission-level requirements Flash LiDAR sensor expertise and interface definition System-level verification and validation
Expected R&R of Korean SME Partner	Definition of processing and computing requirements aligned with mission constraints Development and optimization of processing hardware and/or algorithms Support to system integration and validation activities If possible, act as end user: requirements definition, performances validation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcome:** TRL advancement and mission-ready engineering assets enabling direct integration of Flash LiDAR into future lunar exploration and In-Situ Resource Utilization missions. **Impact:** The proposed Flash LiDAR R&D activity will have a direct and substantial impact on the safety and autonomy of lunar missions. By enabling real-time, full-field 3D perception during terminal descent and surface operations, the proposed Flash LiDAR technology significantly enhances mission safety and autonomy. It reduces operational risks linked to hazardous terrain (e.g. boulders, craters, steep slopes), challenging illumination conditions, and dust disturbances generated during landing and surface activities.

6. References

(1) A. Pollini, C. Pache and J. Haesler, "CSEM Space Lidars for Imaging and Range finding," IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium, DOI: 10.1109/IGARSS.2018.8519241

(2) Roulet, JC., et al. "New Space Compact Flash Lidar", 2024, in Space-based Lidar Remote Sensing Techniques and Emerging Technologies. LIDAR 2023. Springer Aerospace Technology. Springer, Cham. DOI: 10.1007/978-3-031-53618-2_13

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization